

Classroom Grading System

8086 Assembly Language - Microprocessor Course

Item	Description
Project Name	Classroom Grading System
Language	8086 Assembly
Target	MTS-86C 8086 training kit or compatible 8086 environment
Main Task	Sort students in descending order according to their grades
Key Kit Feature	The physical F key works as Enter/Confirm and as Next Result during LCD output

1. Abstract

This project implements a classroom grading system using 8086 Assembly language. The program receives the number of students, then reads student numbers and grades from the keypad. It stores the two groups of data in separate word arrays and sorts them in descending order according to the grades. During sorting, the matching student number is swapped whenever a grade is swapped, so the relationship between each student and his grade is preserved. The final sorted result is printed on the console and shown on the LCD one record at a time.

2. Problem Statement

A classroom contains N students. The student numbers and their grades are stored in memory in two separate tables. The required task is to rearrange the two tables in descending order according to the grades. The output should contain the student numbers arranged according to their grades and the corresponding sorted grades.

3. Objectives

- Read the number of students from the kit keypad.
- Read student numbers and grades as multi-digit numbers.
- Use the physical F key as the Enter/Confirm key for input.
- Store student numbers and grades in two separate arrays.
- Sort grades in descending order.
- Keep every student number linked to its correct grade during sorting.
- Display sorted output through the console and through the LCD.
- Use the F key again to navigate through sorted LCD results one by one.

4. Implemented Features from the Source Code

This section summarizes the actual features included in the submitted assembly source code.

- Keypad input is handled through the KEYPAD procedure.
- READ_NUMBER reads multi-digit numbers by multiplying the previous value by 10 and adding the new digit.
- The physical F key terminates each number input.
- Entered digits are echoed on the LCD while typing.
- The program stores student numbers in STUDENT_NUMS and grades in GRADES.
- Bubble Sort is implemented in SORT_BY_GRADES.
- Sorting is descending by grade.
- Whenever two grades are swapped, the two matching student numbers are also swapped.
- Equal grades keep their original order because the code uses JGE to skip swapping when the current grade is greater than or equal to the next grade.
- The LCD shows one sorted student and grade at a time.
- After showing a sorted record, the program waits in WAIT_F_KEY until F is pressed before showing the next record.
- LCD commands and characters are written using IRWR and OUTL after checking the busy flag using BUSY.

- Console output is also included using INT 21H routines.

5. Important F-Key Behavior

The physical F key is one of the most important practical features of the project. It is used in two different stages: input confirmation and output navigation.

Stage	How F is Used
Input	The user types the digits of a number, then presses F to confirm that number.
Output	After sorting, the LCD displays one result. Pressing F moves to the next sorted result.

In the code, the keypad routine adds 30H to the returned key value. Therefore, the physical F key is detected using CMP AL, 3FH.

6. Program Design

6.1 Data Structures

Name	Declaration / Type	Purpose
N	DW	Stores the number of students after input.
STUDENT_NUMS	DW 100 DUP(?)	Stores up to 100 student numbers.
GRADES	DW 100 DUP(?)	Stores up to 100 grades. Each grade corresponds to the student number at the same index.

6.2 Main Procedures

Procedure	Function
MAIN	Controls the complete program sequence: prompts, input loops, sorting, and output loops.
READ_NUMBER	Builds a multi-digit number from keypad digits until F is pressed.
SORT_BY_GRADES	Sorts the GRADES array in descending order and swaps matching STUDENT_NUMS entries.
PRINT_LCD_NUMBER	Converts a binary word into decimal digits and prints it on the LCD.
PRINT_NUMBER	Converts a binary word into decimal digits and prints it on the console.
KEYPAD	Initializes/reads the keypad interface and returns the pressed key value.
IRWR	Writes LCD instructions such as clear display or move cursor.
OUTL	Writes character data to the LCD.
BUSY	Waits until the LCD is ready before writing.

6.3 Hardware and I/O Addresses

Symbol	Address	Purpose
IR_WR	0FFC1H	LCD instruction register write.
IR_RD	0FFC3H	LCD instruction register read / busy flag.
DR_WR	0FFC5H	LCD data register write.
DATA79	0FFE8H	Keypad data port.
CNTR79	0FFEAH	Keypad control/status port.

7. User Interaction Flow

1. The LCD asks for the number of students.
2. The user enters the digits of N and presses F.
3. The program asks for student numbers. Each student number is typed and confirmed using F.
4. The program asks for grades. Each grade is typed and confirmed using F.
5. The program sorts the arrays by grade in descending order.
6. The first sorted student and grade are displayed on the LCD.
7. If more results remain, the program waits until the user presses F.
8. Every F press shows the next sorted student and grade until the final result is displayed.

8. Sorting Algorithm

Bubble Sort is used because it is simple and suitable for an educational 8086 assembly project.

```
For pass = 0 to N - 2
  For each adjacent pair of grades
    If GRADES[i] < GRADES[i + 1] Then
      Swap GRADES[i] and GRADES[i + 1]
      Swap STUDENT_NUMS[i] and STUDENT_NUMS[i + 1]
    End If
  End For
End For
```

The comparison skips swapping when the current grade is greater than or equal to the next grade. Therefore, equal grades keep their original relative order.

9. Sample Input and Output

The notation 101F means: type 101 on the keypad, then press the physical F key.

Sample input:

```
Number of students: 5F
Student numbers: 101F, 102F, 103F, 104F, 105F
Grades: 75F, 90F, 60F, 85F, 95F
```

Expected sorted output:

```
Student Number: 105, Grade: 95
Student Number: 102, Grade: 90
Student Number: 104, Grade: 85
Student Number: 101, Grade: 75
Student Number: 103, Grade: 60
```

LCD navigation after sorting:

```
LCD shows: Stud#: 105 / Grade: 95
Press F
LCD shows: Stud#: 102 / Grade: 90
Press F
LCD shows: Stud#: 104 / Grade: 85
Press F
LCD shows: Stud#: 101 / Grade: 75
Press F
LCD shows: Stud#: 103 / Grade: 60
```

10. Complexity Analysis

Metric	Value
Time Complexity	$O(N^2)$, because Bubble Sort uses nested passes over adjacent elements.
Memory Complexity	$O(N)$, because the program stores student numbers and grades in two arrays.

11. Implementation Notes and Limitations

- The current array capacity is 100 students.
- The keypad input routine accepts unsigned numeric input only.
- The program does not include validation for invalid grades or values greater than 100.
- The F key is required after every input number and between LCD result screens.
- The source combines kit LCD/keypad I/O with console output using INT 21H, so output behavior depends on the testing environment.
- The LCD number printing routine handles zero values. The console PRINT_NUMBER zero branch should be corrected before using zero in console-based testing.

12. Conclusion

The project demonstrates a complete 8086 assembly solution for sorting classroom grades. It covers memory arrays, loops, conditional jumps, procedures, keypad input, LCD output, and console output. The most important functional point is preserving the link between every student number and its grade while sorting. The kit interaction is also improved by using the F key as both the input confirmation key and the navigation key for viewing sorted LCD results.